LIOUVILLE AND LYAPUNOV CRITERIA FOR ANCIENT NAVIER–STOKES SOLUTIONS

GUILLEM DURAN BALLESTER

Abstract. We study bounded ancient solutions of the three-dimensional Navier–Stokes equations arising from Type I blow-up arguments. We state standard PDE hypotheses under which such solutions are spatially constant; in the Type I class, these constants vanish. We also prove the perturbative weighted vorticity Lyapunov rigidity used by the unconditional Type I chain.

For the unforced equations, known axisymmetric Liouville theorems cover the following cases: absence of swirl, pointwise scale-invariant control, finite $L_t^\infty L_x^p$ control of the swirl variable $\Gamma = ru_\theta$, and axial periodicity with bounded Γ. When one of these hypotheses holds for a nonzero Type I blow-up limit, the Galilean-trivial conclusions of the cited Liouville theorems reduce to the zero solution and contradict the nontriviality of the limit.

We also record the Lyapunov rigidity argument coming from Gallay–Wayne's small-data decay theorem for the three-dimensional rescaled vorticity equation. It yields a coercive Lyapunov functional forcing any ancient trajectory that stays in the perturbative weighted-vorticity ball to be zero. The paper concludes with a summary of the unconditional Liouville hypotheses and cited PDE results used by the chain.

1. Introduction

1.1. Ancient solutions from Type I blow-up. The Type I blow-up argument in Paper I [3] produces, after parabolic rescaling around a Type I singular point, a nonzero ancient solution (U, P) of

$$\partial_t U + (U \cdot \nabla)U + \nabla P = \Delta U, \qquad \nabla \cdot U = 0, \qquad t < 0, \tag{1.1}$$

satisfying the Type I ancient bound

$$\sup_{t<0} \sqrt{-t}\, \|U(t)\|_{L^\infty(\mathbb{R}^3)} < \infty. \tag{1.2}$$

Thus every result proving $U \equiv 0$ for an ancient solution satisfying (1.2) has the same consequence in the Type I problem:

$$\text{PDE hypothesis} \quad \Longrightarrow \quad U \equiv 0 \quad \Longrightarrow \quad \text{contradiction with nonzero blow-up limit.}$$

The paper treats hypotheses for which a specific Liouville mechanism is available. The cases considered are:

$$\text{axisymmetric Liouville hypotheses}, \qquad \text{small weighted rescaled vorticity}.$$

We make no assertion for decaying or symmetry-restricted ancient solutions outside the hypotheses listed below. Each conclusion is tied to an explicit PDE hypothesis and a specified analytic result.

2020 *Mathematics Subject Classification.* 35Q30, 35B40, 35B44, 35B45, 76D05.

Key words and phrases. Navier–Stokes equations, ancient solutions, Type I blow-up, axisymmetric flows, Lyapunov functionals.

For the Type I application in Paper I, only those cases in which the stated PDE hypothesis and the cited or proved Liouville theorem imply $U \equiv 0$ are used as exclusions. Ancient profiles outside these hypotheses are treated separately.

1.2. **Axisymmetric Liouville hypotheses.** Several unforced axisymmetric cases are already covered by known Liouville theorems. Koch–Nadirashvili–Seregin–Šverák cover the no-swirl case and the pointwise scale-invariant condition $|u| \le C/r$. The swirl-bearing finite-L^p condition

$$\Gamma = ru_\theta \in L^\infty_t L^p_x, \qquad 1 \le p < \infty,$$

is covered by Lei–Zhang–Zhao. The z-periodic case with bounded Γ is covered by Lei–Ren–Zhang.

The only additional step needed for Type I ancient limits is to convert Galilean-triviality into zero in the blow-up frame. We time-shift away from the possible singular time $t = 0$, apply the bounded ancient Liouville theorem, patch the resulting constants on backward intervals, and use (1.2): if $U(x,t) \equiv c$, then

$$\sup_{t<0} \sqrt{-t}\, \|U(t)\|_\infty = \sup_{t<0} \sqrt{-t}\, |c|$$

is finite only when $c = 0$.

1.3. **Lyapunov hypotheses.** The Lyapunov result below uses a forward Lyapunov functional in the Gallay–Wayne perturbative weighted-vorticity regime. If w solves the rescaled three-dimensional vorticity equation and remains in the Gallay–Wayne small ball of $L^2_\sigma(m)$, $m > 7/2$, then the functional

$$\mathcal{L}_m(w_0) = \sup_{\tau \ge 0} e^{2\tau} \|\Phi_\tau w_0\|_m^2 \tag{1.3}$$

is coercive and contracts along the forward semiflow. Sending the starting time to $-\infty$ then forces every ancient trajectory in the small ball to be zero.

1.4. **Main results.** The first result records the unforced axisymmetric Liouville consequences relevant to the Type I argument.

Theorem 1.1 (Axisymmetric Liouville consequences for Type I ancient solutions)**.** *Let U be a smooth Type I ancient solution of (1.1) satisfying (1.2). If U satisfies any of the following axisymmetric assumptions, then $U \equiv 0$:*

(i) *axisymmetric without swirl;*

(ii) *axisymmetric and satisfying a pointwise scale-invariant bound*

$$|U(x,t)| \le \frac{C}{r};$$

(iii) *axisymmetric with*

$$\Gamma = rU_\theta \in L^\infty_t L^p_x\big(\mathbb{R}^3 \times (-\infty, 0)\big), \qquad 1 \le p < \infty;$$

(iv) *axisymmetric, periodic in the physical axial variable z, and with*

$$\Gamma = rU_\theta \in L^\infty\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$

Consequently none of these assumptions can hold for a nonzero Type I blow-up limit.

Theorem 1.2 (Perturbative weighted-vorticity Lyapunov rigidity)**.** *Let $m > \frac{7}{2}$. Let $w(\tau)$ be an ancient mild solution of the Gallay–Wayne rescaled vorticity equation*

$$\partial_\tau w = \Lambda w - (v \cdot \nabla) w + (w \cdot \nabla) v, \qquad \nabla \cdot w = 0, \tag{1.4}$$

with v recovered by the three-dimensional Biot–Savart law. Assume

$$\sup_{\tau \in \mathbb{R}} \|w(\tau)\|_{L^2_\sigma(m)} \leq \rho_m,$$

where ρ_m is the Gallay–Wayne small-data radius. Then

$$w \equiv 0.$$

Here "ancient mild" means that every forward time-shift of w is the Gallay–Wayne mild solution launched from its value at the starting time.

Theorem 1.3 (Consequences for Type I blow-up limits)**.** *Let U be a nonzero Type I blow-up limit. Then U cannot satisfy any of the axisymmetric assumptions in Theorem 1.1; nor can its corresponding rescaled vorticity satisfy the hypotheses of Theorem 1.2.*

Only the explicit hypotheses stated above are used.

1.5. **Relation to Papers I and II.** Paper I [3] reduces Type I singularities to nonzero ancient limits satisfying (1.2). Paper II [4] treats uniformly L^3-tight ancient solutions. The present paper records the unconditional symmetry and perturbative weighted-vorticity exclusions used by the chain. Known Liouville theorems rule out the axisymmetric cases listed above, and the Lyapunov argument handles the perturbative weighted-vorticity case. Paper III does not depend on Paper II.

1.6. **Summary of hypotheses.**

Hypothesis	Conclusion	Analytic ingredient
axisymmetric no swirl	$U \equiv 0$	KNSS
pointwise $\|u\| \leq C/r$	$U \equiv 0$	KNSS
$\Gamma \in L^\infty_t L^p_x$, $1 \leq p < \infty$	$U \equiv 0$	LZZ
z-periodic bounded Γ	$U \equiv 0$	LRZ
small weighted vorticity	$w \equiv 0$	Gallay–Wayne

1.7. **Organization.** Section 2 records the Type I ancient setup, self-similar variables, the invariance facts, and the time-shift lemma. Section 3 fixes axisymmetric notation and the known Liouville theorems. Section 4 proves the unforced axisymmetric Liouville consequences. Section 5 proves the Gallay–Wayne Lyapunov rigidity, and Section 6 summarizes the consequences.

2. Ancient Type I limits and self-similar variables

2.1. Type I ancient solutions.

Definition 2.1 (Type I ancient solution)**.** A smooth mild solution U of (1.1) on $\mathbb{R}^3 \times (-\infty, 0)$ is called Type I ancient if (1.2) holds. Mild is understood in the standard velocity formulation [7, 14]: for every $-\infty < s < t < 0$,

$$U(t) = e^{(t-s)\Delta} U(s) - \int_s^t e^{(t-\sigma)\Delta} \mathbb{P} \nabla \cdot (U(\sigma) \otimes U(\sigma))\, d\sigma \tag{2.1}$$

in distributions, where $\mathbb{P}$ is the Leray projector.

2.2. **Nonzero Type I blow-up limits.**

Definition 2.2 (Nonzero Type I blow-up limit)**.** A Type I ancient solution U is called a nonzero Type I blow-up limit if it arises as the limit of parabolic rescalings around a Type I singular point and the limiting solution is not identically zero. The arguments below require only the conclusion $U \not\equiv 0$.

Lemma 2.3 (Nontriviality of blow-up limits)**.** *A nonzero Type I blow-up limit is not identically zero.*

Proof. This is part of the definition above. ☐

2.3. **Backward self-similar variables.** For $t < 0$, define

$$V(y,\tau) = \sqrt{-t}\,U(x,t), \qquad Q(y,\tau) = (-t)P(x,t), \qquad y = \frac{x}{\sqrt{-t}}, \qquad \tau = -\log(-t). \tag{2.2}$$

Equivalently,

$$x = e^{-\tau/2}y, \qquad t = -e^{-\tau}.$$

Lemma 2.4 (Self-similar equation)**.** *Let (U,P) be a smooth solution of (1.1), and define (V,Q) by (2.2). Then V is defined on $\mathbb{R}^3 \times \mathbb{R}$ and satisfies*

$$\partial_\tau V + (V\cdot\nabla)V + \nabla Q = \Delta V - \frac{1}{2}(V + y\cdot\nabla V), \qquad \nabla\cdot V = 0. \tag{2.3}$$

Moreover (1.2) is equivalent to

$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

Proof. Let $a = \sqrt{-t} = e^{-\tau/2}$. Then $V = aU$, $y = x/a$, and

$$\partial_t\tau = \frac{1}{-t} = \frac{1}{a^2}, \qquad \partial_t y = \frac{y}{2a^2}, \qquad \partial_t(a^{-1}) = \frac{1}{2a^3}.$$

Since $U = a^{-1}V$, differentiating at fixed x gives

$$\partial_t U = a^{-3}\left(\partial_\tau V + \frac{1}{2}y\cdot\nabla V + \frac{1}{2}V\right).$$

Spatial derivatives scale as

$$\nabla_x U = a^{-2}\nabla_y V, \qquad \Delta_x U = a^{-3}\Delta_y V,$$

and the pressure scaling gives

$$\nabla_x P = a^{-3}\nabla_y Q.$$

Thus

$$(U\cdot\nabla_x)U = a^{-3}(V\cdot\nabla_y)V.$$

Multiplying (1.1) by a^3 yields (2.3); the divergence equation follows from $\nabla_x\cdot U = a^{-2}\nabla_y\cdot V$. The spatial map $y \mapsto x = e^{-\tau/2}y$ is a bijection for each τ, and

$$\|V(\tau)\|_\infty = \sqrt{-t}\,\|U(t)\|_\infty.$$

Taking suprema over $t < 0$, equivalently $\tau\in\mathbb{R}$, gives the last assertion. ☐

2.4. **Preservation of axisymmetry and swirl variables.** For axisymmetric flows we write

$$x = (r\cos\theta, r\sin\theta, z), \qquad U = U_r e_r + U_\theta e_\theta + U_z e_z,$$

and in centered variables

$$y = (\rho\cos\theta, \rho\sin\theta, Z), \qquad V = V_\rho e_\rho + V_\theta e_\theta + V_Z e_Z.$$

The scalar swirl variables are

$$\Gamma_U(r,z,t) = rU_\theta(r,z,t), \qquad \Gamma_V(\rho,Z,\tau) = \rho V_\theta(\rho,Z,\tau).$$

Lemma 2.5 (Preservation of axisymmetric structure)**.** *Axisymmetry and absence of swirl are preserved under* (2.2)*. If U is axisymmetric, then*

$$\Gamma_V(\rho,Z,\tau) = \Gamma_U(r,z,t), \qquad (r,z) = \sqrt{-t}\,(\rho,Z).$$

Fixed physical z-periodicity is preserved by physical time shifts, but under the centered change of variables it becomes a τ-dependent period in the Z-coordinate.

Proof. The self-similar map is a scalar spatial dilation and a scalar multiplication of the velocity, so it commutes with rotations around the symmetry axis. The cylindrical basis is unchanged by radial dilation, and

$$V_\theta(\rho,Z,\tau) = \sqrt{-t}\,U_\theta(r,z,t), \qquad r = \sqrt{-t}\rho, \quad z = \sqrt{-t}Z.$$

Thus $U_\theta \equiv 0$ if and only if $V_\theta \equiv 0$, and

$$\Gamma_V = \rho V_\theta = \rho\sqrt{-t}\,U_\theta = rU_\theta = \Gamma_U.$$

If U has physical axial period L, then V has period $L/\sqrt{-t} = Le^{\tau/2}$ in Z, not a fixed period. □

2.5. **Time shifts away from the singular time.**

Lemma 2.6 (Time shifts away from the singular time)**.** *Let U be a smooth Type I ancient solution and fix $T < 0$. Then*

$$W_T(x,s) = U(x,T+s), \qquad \Pi_T(x,s) = P(x,T+s), \qquad s < 0,$$

is a smooth bounded mild ancient solution of (1.1) *on $\mathbb{R}^3 \times (-\infty,0)$. Moreover the axisymmetric, no-swirl, L^p-swirl, bounded-Γ, and physical z-periodic properties are preserved.*

Proof. Time translation preserves the differential equation because $\partial_s W_T(x,s) = \partial_t U(x,T+s)$ and all spatial derivatives are unchanged. For the mild formulation, apply (2.1) to U between the physical times $T+s < T+t < 0$:

$$U(T+t) = e^{(t-s)\Delta}U(T+s) - \int_{T+s}^{T+t} e^{(T+t-\eta)\Delta}\mathbb{P}\nabla\cdot(U(\eta)\otimes U(\eta))\,d\eta.$$

With $\eta = T+\sigma$, this becomes the mild formula for W_T on the interval (s,t). For $s < 0$, $T+s < T < 0$, so the Type I bound gives

$$\|W_T(s)\|_\infty = \|U(T+s)\|_\infty \le \frac{1}{\sqrt{-T}} \sup_{\eta<0} \sqrt{-\eta}\,\|U(\eta)\|_\infty.$$

The invariance assertions follow because only the time variable has been translated. In particular

$$\Gamma_{W_T}(r,z,s) = \Gamma_U(r,z,T+s),$$

so $L^\infty_s L^p_x$ and L^∞ bounds for Γ restrict to the shifted solution, and a fixed physical axial period remains the same. □

2.6. **PDE hypotheses used below.** The arguments below are stated directly in terms of hypotheses on ancient solutions. The unconditional axisymmetric hypotheses are:

(i) axisymmetry and $U_\theta \equiv 0$;
(ii) axisymmetry and the pointwise bound $|U(x,t)| \le C/r$ for $r > 0$;
(iii) axisymmetry and

$$\Gamma = rU_\theta \in L_t^\infty L_x^p\big(\mathbb{R}^3 \times (-\infty,0)\big), \qquad 1 \le p < \infty;$$

(iv) axisymmetry, periodicity in the physical axial variable z, and $\Gamma \in L^\infty$.

The additional hypothesis used later is the Gallay–Wayne small weighted-vorticity condition.

3. Axisymmetric notation and known Liouville theorems

3.1. **Cylindrical variables and the swirl scalar.** For an axisymmetric solution u, write

$$u = u_r e_r + u_\theta e_\theta + u_z e_z, \qquad \Gamma = ru_\theta, \qquad b = u_r e_r + u_z e_z.$$

When Γ is placed in $L^p(\mathbb{R}^3)$, it is regarded as the corresponding function of $x = (r\cos\theta, r\sin\theta, z)$, independent of θ:

$$\|\Gamma(t)\|_{L^p(\mathbb{R}^3)}^p = 2\pi \int_{\mathbb{R}} \int_0^\infty |\Gamma(r,z,t)|^p r\, dr\, dz, \qquad 1 \le p < \infty.$$

3.2. **Constant and Galilean-trivial solutions.**

Definition 3.1 (Galilean-trivial)**.** A velocity field is Galilean-trivial if, after a Galilean transform, it is a constant vector field. In the Type I blow-up frame below we use only the following consequence: when a cited Liouville theorem gives a constant velocity field in that frame, the Type I bound forces the constant to be zero.

Lemma 3.2 (Spatially constant mild solutions)**.** *Let W be a smooth mild solution of* (1.1) *on* $\mathbb{R}^3 \times (-\infty,0)$*. If $W(x,t) = b(t)$ for all x and t, then b is constant in time.*

Proof. Insert $W(x,t) = b(t)$ into (2.1). The heat semigroup leaves constant vector fields fixed, and $\nabla \cdot (b(\sigma) \otimes b(\sigma)) = 0$. Thus $b(t) = b(s)$ for every $s < t < 0$. □

Lemma 3.3 (Patching constants on backward intervals)**.** *Let $U : \mathbb{R}^3 \times (-\infty,0) \to \mathbb{R}^3$ be a function. Assume that for every $T < 0$ there exists $c_T \in \mathbb{R}^3$ such that*

$$U(x,t) = c_T \qquad (x \in \mathbb{R}^3,\ t < T).$$

Then there is a single $c \in \mathbb{R}^3$ such that

$$U(x,t) = c \qquad (x \in \mathbb{R}^3,\ t < 0).$$

Proof. If $T_1 < T_2 < 0$, the two constants agree on the nonempty interval $(-\infty, T_1)$. Hence all c_T are equal. For any $t < 0$, choose T with $t < T < 0$. □

Lemma 3.4 (Constant Type I ancient solutions vanish)**.** *Let U be a Type I ancient solution. If $U(x,t) \equiv c$, then $c = 0$.*

Proof. The Type I bound gives

$$\sqrt{-t}\,|c| \le \sup_{s<0} \sqrt{-s}\,\|U(s)\|_\infty < \infty \qquad (t < 0).$$

Letting $t \to -\infty$ gives $c = 0$. □

3.3. **Known Liouville theorems.** We use the following Liouville theorems in the stated forms.

Theorem 3.5 (KNSS no-swirl Liouville theorem)**.** *Every bounded ancient mild axisymmetric no-swirl solution of the unforced Navier–Stokes equations is Galilean-trivial. In the form used below, the solution is a spatially constant axial drift.*

Proof. This is the no-swirl Liouville theorem of Koch–Nadirashvili–Seregin–Šverák [8], stated in the form needed for the Type I vanishing argument. □

Theorem 3.6 (KNSS pointwise scale-invariant condition)**.** *Every bounded ancient mild axisymmetric solution satisfying*

$$|u(x,t)| \le \frac{C}{r} \qquad (r > 0,\ t < 0)$$

is Galilean-trivial. In the form used below, it is identically zero.

Proof. This is the pointwise Liouville theorem of [8], stated in the form used here. □

Theorem 3.7 (Lei–Zhang–Zhao)**.** *Let u be a bounded ancient mild axisymmetric solution. If*

$$\Gamma = ru_\theta \in L^\infty_t L^p_x\big(\mathbb{R}^3 \times (-\infty, 0)\big), \qquad 1 \le p < \infty,$$

then u is a constant vector field.

Proof. This is the ancient axisymmetric L^p-swirl Liouville theorem of Lei–Zhang–Zhao [11]. □

Theorem 3.8 (Lei–Ren–Zhang)**.** *Let u be a bounded ancient mild axisymmetric solution, periodic in z, with*

$$\Gamma = ru_\theta \in L^\infty\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$

Then u is a constant axial vector field.

Proof. This is the periodic bounded-swirl theorem of Lei–Ren–Zhang [10]. Their statement is on $\mathbb{R}^2 \times \mathbb{T}^1$; a physical z-periodic whole-space solution descends to the quotient, and a Navier–Stokes scaling normalizes the period. □

4. Unforced axisymmetric Liouville consequences

4.1. No swirl.

Proposition 4.1 (No-swirl Type I vanishing)**.** *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and $U_\theta \equiv 0$. Then $U \equiv 0$.*

Proof. Fix $T < 0$ and set $W_T(x,s) = U(x, T+s)$. By Theorem 2.6, W_T is a bounded ancient mild axisymmetric no-swirl solution. Applying Theorem 3.5 gives that W_T is a spatially constant axial drift, say $W_T(x,s) = b_T(s)e_z$. By Theorem 3.2, $b_T(s)$ is constant in s. Thus $U(x,t) = c_T$ for $t < T$. Since $T < 0$ is arbitrary, Theorem 3.3 gives $U(x,t) = c$ for all $t < 0$. Finally, Theorem 3.4 gives $c = 0$. □

4.2. Pointwise scale-invariant control.

Proposition 4.2 (Pointwise $1/r$ Type I vanishing)**.** *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and satisfies*

$$|U(x,t)| \le \frac{C}{r} \qquad (r > 0,\ t < 0).$$

Then $U \equiv 0$.

Proof. Fix $T < 0$. The shifted solution $W_T(x,s) = U(x, T+s)$ is bounded, ancient, axisymmetric, and satisfies the same pointwise bound $|W_T| \le C/r$. Theorem 3.6 gives $W_T \equiv 0$. Hence $U \equiv 0$ on $(-\infty, T)$. Since $T < 0$ is arbitrary, $U \equiv 0$. ☐

4.3. Finite L^p control of Γ.

Proposition 4.3 (Finite-L^p swirl Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and, for some $1 \le p < \infty$,*

$$\Gamma = rU_\theta \in L^\infty_t L^p_x\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$

Then $U \equiv 0$.

Proof. Fix $T < 0$. By Theorem 2.6, $W_T(x,s) = U(x, T+s)$ is a bounded ancient mild axisymmetric solution and

$$\Gamma_{W_T} \in L^\infty_s L^p_x\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$

Theorem 3.7 gives that W_T is a constant vector field. Patch these constants in T by Theorem 3.3 and then use Theorem 3.4. ☐

4.4. Periodic bounded swirl.

Proposition 4.4 (Periodic bounded-swirl Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric, periodic in the physical axial variable z, and satisfies*

$$\Gamma = rU_\theta \in L^\infty\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$

Then $U \equiv 0$.

Proof. Fix $T < 0$. By Theorem 2.6, the shifted solution W_T is bounded, ancient, axisymmetric, physically z-periodic, and has bounded Γ. Theorem 3.8 gives that W_T is a constant axial vector field. The constants patch as T varies, and the Type I bound eliminates the resulting constant by Theorem 3.4. ☐

4.5. Proof of the axisymmetric Liouville theorem.

Proof of Theorem 1.1. The four cases are precisely the hypotheses of Theorems 4.1 to 4.4. Each proposition gives $U \equiv 0$. This contradicts Theorem 2.3 when U is a nonzero Type I blow-up limit. ☐

5. Perturbative weighted-vorticity Lyapunov rigidity

5.1. The rescaled vorticity equation. In this section τ is the forward self-similar time used in the Gallay–Wayne vorticity formulation. The ancient trajectories are indexed by all $\tau \in \mathbb{R}$, and the forward semiflow acts by increasing τ.

Let $w = \nabla \times v$. The rescaled vorticity equation is

$$\partial_\tau w = \Lambda w - (v \cdot \nabla) w + (w \cdot \nabla) v, \qquad \nabla \cdot w = 0, \tag{5.1}$$

where

$$\Lambda w = \Delta w + \frac{1}{2}\xi \cdot \nabla w + w, \tag{5.2}$$

and v is recovered from w by the three-dimensional Biot–Savart law. For $m \ge 0$, set

$$L^2_\sigma(m) = \left\{ f \in L^2(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot f = 0, \quad \|f\|_m < \infty \right\},$$

where

$$\|f\|_m^2 = \int_{\mathbb{R}^3} (1 + |\xi|)^{2m} |f(\xi)|^2 \, d\xi.$$

5.2. **Gallay–Wayne small-data decay.**

Theorem 5.1 (Gallay–Wayne small-data decay). *Let $0 < \mu \le 1$ and $m > 2\mu + \frac{1}{2}$. There exist constants $\rho_m > 0$ and $C_m \ge 1$ such that if $\|w_0\|_m \le \rho_m$, then (5.1) has a unique global mild solution $w(\tau) = \Phi_\tau w_0$ in $C([0,\infty); L^2_\sigma(m))$ and*

$$\|\Phi_\tau w_0\|_m \le C_m e^{-\mu\tau}\|w_0\|_m, \qquad \tau \ge 0.$$

In particular, if $m > \frac{7}{2}$, one may use $\mu = 1$.

Proof. This is the Gallay–Wayne weighted small-data theorem for the rescaled three-dimensional vorticity equation [6]. The only conclusions used below are existence, uniqueness, the semiflow property, and the displayed exponential decay. □

5.3. **The Lyapunov functional.**

Definition 5.2 (Perturbative weighted-vorticity condition). Fix $m > \frac{7}{2}$, and let ρ_m be the radius in Theorem 5.1. An ancient vorticity trajectory satisfies the perturbative weighted-vorticity condition with exponent m if it is a function

$$w \in C(\mathbb{R}; L^2_\sigma(m))$$

that solves (5.1) for all $\tau \in \mathbb{R}$ and satisfies

$$\sup_{\tau\in\mathbb{R}} \|w(\tau)\|_m \le \rho_m.$$

Solving for all τ means that every time shift is compatible with the Gallay–Wayne forward mild semiflow.

Proposition 5.3 (Semiflow Lyapunov functional). *Fix $m > \frac{7}{2}$. For $\|w_0\|_m \le \rho_m$, define*

$$\mathcal{L}_m(w_0) = \sup_{\tau\ge 0} e^{2\tau}\|\Phi_\tau w_0\|_m^2. \tag{5.3}$$

Then $\mathcal{L}_m$ is well-defined and satisfies

$$\|w_0\|_m^2 \le \mathcal{L}_m(w_0) \le C_m^2\|w_0\|_m^2. \tag{5.4}$$

Moreover, for every $s \ge 0$,

$$\mathcal{L}_m(\Phi_s w_0) \le e^{-2s}\mathcal{L}_m(w_0). \tag{5.5}$$

Proof. The Gallay–Wayne estimate with $\mu = 1$ gives

$$e^{2\tau}\|\Phi_\tau w_0\|_m^2 \le C_m^2\|w_0\|_m^2,$$

which proves finiteness and the upper bound. The lower bound is the value at $\tau = 0$. By uniqueness,

$$\Phi_\tau(\Phi_s w_0) = \Phi_{\tau+s} w_0.$$

Hence

$$\mathcal{L}_m(\Phi_s w_0) = \sup_{\tau\ge 0} e^{2\tau}\|\Phi_{\tau+s} w_0\|_m^2 = e^{-2s}\sup_{\sigma\ge s} e^{2\sigma}\|\Phi_\sigma w_0\|_m^2 \le e^{-2s}\mathcal{L}_m(w_0).$$

□

5.4. **Ancient rigidity in the small weighted ball.**

Proof of Theorem 1.2. Let w be an ancient solution in the small ball and fix $\tau_0 \in \mathbb{R}$. For $s < \tau_0$, the shifted function

$$W_s(\theta) = w(s+\theta), \qquad \theta \geq 0,$$

is the Gallay–Wayne mild solution with initial datum $w(s)$. The small-ball hypothesis gives $\|w(s)\|_m \leq \rho_m$, so the forward solution is in the domain of Φ_θ, and uniqueness gives

$$W_s(\theta) = \Phi_\theta w(s), \qquad \theta \geq 0.$$

Choosing $\theta = \tau_0 - s$ gives

$$w(\tau_0) = \Phi_{\tau_0 - s} w(s).$$

Since $\|w(\tau_0)\|_m \leq \rho_m$, the Lyapunov functional is also defined at $w(\tau_0)$. Applying Theorem 5.3 with forward time $\tau_0 - s$,

$$\mathcal{L}_m(w(\tau_0)) \leq e^{-2(\tau_0 - s)} \mathcal{L}_m(w(s)) \leq C_m^2 \rho_m^2 e^{-2(\tau_0 - s)}.$$

Letting $s \to -\infty$ gives $\mathcal{L}_m(w(\tau_0)) = 0$. The coercive lower bound (5.4) gives $w(\tau_0) = 0$. Since τ_0 was arbitrary, $w \equiv 0$. □

6. Summary of PDE consequences

6.1. **Zero conclusions.**

Theorem 6.1 (Zero conclusions for Type I blow-up limits)**.** *The following hypotheses cannot hold for a nonzero Type I blow-up limit:*

(i) *axisymmetry and $U_\theta \equiv 0$;*
(ii) *axisymmetry and the pointwise bound $|U(x,t)| \leq C/r$;*
(iii) *axisymmetry and finite-L^p control of the swirl scalar,*

$$\Gamma \in L_t^\infty L_x^p, \qquad 1 \leq p < \infty;$$

(iv) *axisymmetry, z-periodicity, and bounded Γ;*
(v) *the Gallay–Wayne small weighted-vorticity condition, whenever the velocity is recovered from the corresponding vorticity by the Biot–Savart law.*

Proof. Items (i)–(iv) are Theorem 1.1. Item (v) is Theorem 1.2; since the velocity in that case is recovered from w by the Biot–Savart law, $w \equiv 0$ gives the zero velocity. In each Type I case, the zero conclusion contradicts Theorem 2.3. □

6.2. **Consequences for Type I blow-up arguments.**

Proof of Theorem 1.3. The axisymmetric cases are ruled out by Theorem 1.1. The perturbative weighted-vorticity case is ruled out by Theorem 1.2 once the corresponding rescaled vorticity satisfies the small weighted condition; the Biot–Savart reconstruction then gives zero velocity. These zero conclusions contradict the nontriviality of a Type I blow-up limit. □

Theorem 6.2 (Combined zero conclusion)**.** *Let U be a Type I ancient solution obtained as a nonzero Type I blow-up limit. If U satisfies one of the hypotheses in Theorem 6.1, then $U \equiv 0$, a contradiction.*

Proof. This is Theorem 6.1. □

6.3. Summary table.

Hypothesis	Conclusion	Analytic ingredient
no swirl	$U \equiv 0$	KNSS plus Type I constant removal
pointwise $1/r$	$U \equiv 0$	KNSS
$\Gamma \in L^\infty_t L^p_x$	$U \equiv 0$	LZZ plus Type I constant removal
periodic bounded Γ	$U \equiv 0$	LRZ plus Type I constant removal
small weighted vorticity	$w \equiv 0$	Gallay–Wayne Lyapunov functional

The no-swirl, pointwise scale-invariant, finite-L^p-swirl, and periodic bounded-swirl cases are covered by known axisymmetric Liouville theorems. The perturbative weighted-vorticity case is covered by a Lyapunov functional derived from Gallay–Wayne decay. Thus every zero conclusion used in the Type I application rests on an explicit PDE hypothesis stated in this paper.

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Email address: `guillem@fragile.tech`